Satyam Awasthi

Researcher & Software Engineer

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Professional Summary

Computer Science researcher focused on **Embodied AI**, **Autonomous Systems**, and **Spatial Perception**. My research investigates how perception, planning, and learning-based decision-making can be integrated to enable robust and reliable long-horizon autonomy in robotic systems.

Research Interests

Embodied AI, Autonomous Systems, Spatial Perception, and Learning-Based Control, with emphasis on long-horizon planning and decision-making under uncertainty.

Education

University of California, Santa Barbara (UCSB) — Santa Barbara, CA
M.S. in Computer Science (GPA: 3.91/4.00) — Sep 2021 – Jun 2023

Indian Institute of Technology, Kharagpur (IIT Kharagpur) — Kharagpur, India
B.Tech. in Electrical Engineering, Minor in Computer Science (GPA: 8.98/10.0) — Jul 2016 – Jul 2020

Research Experience

Four Eyes Laboratory, UC Santa Barbara — Sep 2021 – Jun 2023
Graduate Researcher — Advisor: Prof. Tobias H"ollerer, Co-Advisor: Prof. Michael Beyeler

- Developed EyeTTS, a calibration and evaluation framework for eye tracking in mixed-reality locomotion.
- Designed experimental protocols and software pipelines to quantify gaze-tracking drift during active user motion.
- Curated a multi-device eye-tracking dataset and conducted user studies across diverse locomotion layouts.
- Published first-author poster papers at IEEE VR and IEEE ISMAR Adjunct venues.

Autonomous Ground Vehicle & Swarm Robotics Group, IIT Kharagpur — Jan 2016 – Dec 2018
Undergraduate Researcher

- Developed a hardware-agnostic navigation stack integrating EKF-based localization from LiDAR, IMU, and GPS streams.
- Implemented visual obstacle segmentation using a custom U-Net architecture and hybrid motion planning using A*/Dijkstra and Dynamic Window Approach.
- Designed decentralized multi-agent coordination using relative-pose estimation over an ad-hoc mesh network.
- Enabled zero-shot Sim2Real transfer across heterogeneous robotic platforms.

Publications

Peer-Reviewed Conference Papers

SmartWatch for Wall Writing: Real-time Transcription of Wall Writing from Inertial Sensing
Snigdha Das, **Satyam Awasthi**, Abdul Shamnar P, Pradipta De, Sandip Chakraborty, and Bivas Mitra.
COMSNETS, 2022. DOI: 10.1109/COMSNETS53615.2022.9668531

Workshop & Poster Publications

Eye Tracking Performance in Mobile Mixed Reality
Satyam Awasthi, Vivian Ross, Sydney Lim, Michael Beyeler, and Tobias H"ollerer.
IEEE VRW, 2024. DOI: 10.1109/VRW62533.2024.00321

EyeTTS: Evaluating and Calibrating Eye Tracking for Mixed-Reality Locomotion
Satyam Awasthi, Vivian Ross, Michael Beyeler, and Tobias H"ollerer.
ISMAR Adjunct, 2023. DOI: 10.1109/ISMAR-Adjunct60411.2023.00104

Research Artifacts

EyeTTS Calibration Framework 2022–2024 `Calibration Framework` · `User Study Framework`

- Developed an extensible framework for evaluating and calibrating eye-tracking systems during mixed-reality locomotion.
- Implemented device-agnostic benchmarking pipelines and automated drift-correction procedures.
- Released reproducible software infrastructure supporting eye-tracking research.

EyeTTS Mixed-Reality Eye Tracking Dataset 2022–2024 `Dataset` · `User Study Videos`

- Curated a multi-device mixed-reality locomotion dataset for calibration and performance evaluation.
- Collected gaze trajectories, calibration measurements, and locomotion metadata across diverse MR layouts.
- Supported experimental validation reported in ISMAR and VRW publications.

Industry Experience

Intuit Inc Mountain View, CA · Mar 2024 – Jan 2026
Software Engineer 2

- Built dual-stage **agentic decision pipelines** using tool-schema routing for real-time intent classification and sequential state tracking.
- Developed a decoupled **generative UI engine** leveraging RAG and structured LLMs to optimize deterministic component layouts.

Yahoo Inc Mountain View, CA · Apr 2023 – Nov 2023
Software Dev Engineer II (IC3)

- Developed **client-side rule engines** for asynchronous receipt payloads, executing deterministic filtering for **real-time decision support**.

Coupang Global LLC Mountain View, CA · Jun 2022 – Sep 2022
Software Engineering Intern

- Engineered **context-aware heuristics** to prune multi-dimensional search spaces based on active user state vectors.
- Designed **Content-Based Filtering (CBF)** layers using **cosine similarity** to rank high-dimensional product features.

Samsung Research Institute Noida, India · Sep 2020 – Sep 2021
Software Engineer (CL 2)

- Architected a **continuous on-device perception pipeline** ingesting multi-modal media streams for automated ML classification.
- Designed a **cross-profile data brokering engine** leveraging multi-user storage isolation to route assets from **User 1** to sandboxed **User 0**.

Selected Projects

JitterNot Jan 2022 – Mar 2022
Learning-Based Model Predictive Control for Adaptive Video Streaming (UCSB)

- Developed an LSTM-based predictor for network throughput enabling closed-loop control under non-stationary dynamics.
- Designed a model predictive control (MPC) framework using receding horizon optimization for adaptive decision-making over bandwidth constraints.

In Motion — HAR using DeepConvLSTM Nov 2019
Embedded Sensor Inference System for Temporal Activity Recognition (IIT Kharagpur)

- Designed a streaming inference pipeline for high-frequency IMU signals enabling real-time temporal classification.
- Implemented a DeepConvLSTM architecture combining convolutional feature extraction with temporal sequence modeling for activity recognition.

CohesiveAR Apr 2022 – Jun 2022
Spatial Perception Pipeline for Geometry Alignment and View-Invariant Mapping (UCSB)

- Developed a spatial perception pipeline using ARCore and OpenCV for projecting dynamic 3D scene geometry into consistent 2D reference frames.
- Implemented homography-based geometric correction methods to achieve view-invariant surface texture extraction under camera motion.

Tweeter: Distributed Systems Testbed Sep 2021 – Dec 2021
State-Driven Event Processing System for High-Concurrency Workload Modeling (UCSB)

- Modeled user interactions as state machines to analyze system latency scaling under high-concurrency workloads.
- Designed caching and query optimization strategies to improve throughput in distributed database-backed systems under load.

Teaching

University of California, Santa Barbara Santa Barbara, CA
Graduate Teaching Assistant Sep 2021 – Jun 2023

- Directed instructional labs, formulated algorithmic programming assignments, and monitored system architecture milestones for senior capstone teams.
- Covered advanced and core tracks: CS189 (Capstone), CS184 (Android), CS24 (C++ Data Structures), and CS16 (Problem Solving).

Technical Skills

Programming: Python, C++, Java, Kotlin, MATLAB

Robotics & Simulation: ROS/ROS2, Gazebo, RViz, Unity

Machine Learning & Vision: PyTorch, OpenCV

Methods: State Estimation (EKF), Model Predictive Control (MPC), Motion Planning, Sensor Fusion, Spatial Perception

Developer Tools: Git

Academic Service & Peer Review

Peer Reviewer, UIST 2024
ACM Symposium on User Interface Software and Technology

- Reviewed submissions on interactive systems, spatial tracking, and interface hardware.